Structured summary: mean moderation and covariance moderation as data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials (supplementary key-points summary)

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Contents

This document is a structured, section-by-section summary of the key points argued in the companion manuscript, "Mean moderation and covariance moderation as data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials, and their divergent power under carryover." It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.

1 1. Introduction

Power for these designs is estimated by simulation, and the simulation must specify the biomarker interaction in its DGP in more than one possible way.

- The target of inference is the biomarker-treatment interaction.
- Closed-form power exists only in simplified cases [@pmsimstats-paper08]. For the designs and model considered here it is obtained by Monte Carlo simulation.
- How the interaction is built into the DGP is a modeling choice, and that choice is the subject of this paper.

There are two architectures, mean moderation and differential correlation.

- Under mean moderation an interaction term scales the treatment effect in the conditional mean. This is standard across trial-simulation frameworks.
- Under covariance moderation the biomarker changes the response-outcome correlation, so the interaction is represented in the joint distribution. This is the approach taken by Hendrickson et al. for N-of-1 trials.
- A dual-channel architecture activating both at once is developed in a companion paper (Section 2.2).

Hendrickson et al. chose covariance moderation for the prazosin and PTSD trial as the biologically appropriate choice, departing from the modal mean-moderation convention. This paper quantifies what that choice costs under carryover.

- Covariance moderation was the logical specification given the trial's own results, because mean moderation would have imposed a deterministic biomarker-response relationship the biology did not support.
- Mean moderation remains the modal specification in the N-of-1 and broader biomarker-simulation literature, so this was a deliberate departure, not an oversight.
- The size of the resulting power penalty under carryover had not previously been quantified. Doing so is this paper's central aim.

Aggregated N-of-1 trials pool repeated within-patient on-drug and off-drug contrasts, and carryover is the nuisance that distinguishes them.

- A single N-of-1 trial repeatedly crosses one patient between treatment states, and pooling many such series yields population-level inference at small N.
- Carryover is residual drug effect that decays over days to weeks and contaminates off-drug occasions.
- Three within-subject designs, namely crossover (CO), open-label with blinded discontinuation (OL+BDC), and Hybrid, differ in how on-drug and off-drug occasions are arranged.

2 2. Two DGP architectures

Notation. B_i is the biomarker, D_{it} the on-drug indicator, BR_{it} the biological response, and `Dbc` the analysis-layer exposure.

- B_i, D_{it}, BR_{it} describe the data-generating layer.
- The analysis layer uses a continuous exposure `Dbc` (Section 3.1).
- Typewriter identifiers denote code and formula variables. Mathematical symbols are used otherwise.

2.1 2.1 Mean moderation

Under mean moderation the interaction is explicit in the conditional mean, so that a higher biomarker value gives a proportionally larger treatment effect.

- β_{bm} is the biomarker moderation parameter on the centered biomarker.
- The effect scales with B_i deterministically in the conditional mean.
- The signal lives in the first moment of the outcome.

The multiplicative form maps onto the standard additive effect-modeling regression, so both are instances of mean moderation.

- The additive interaction β_3 equals $\beta_1 \cdot \beta_{bm}$ in the multiplicative form.
- The additive form adds a prognostic main effect β_2 the multiplicative form sets to zero.
- For interaction power the two are interchangeable up to that prognostic distinction.

2.2 2.2 Covariance moderation through differential correlation

Under covariance moderation the interaction is in the second moment. The biomarker modulates BR through a correlation that decays during the off-drug period.

- The conditional mean of BR rises with b through the correlation, not through a mean parameter.
- The effective correlation is c_{bm} on drug and $c_{bm}e^{-\lambda t_{sd}}$ off drug.
- Moderation persists with attenuated strength off drug and vanishes only at complete decay.

3 3. Consequences for Statistical Power Under Carryover

3.1 3.1 Empirical demonstration

Under mean moderation the carryover-induced power loss is modest in the two-period designs (2-3% relative) and larger in the Hybrid design (7%).

- Relative loss to $t_{1/2} = 1$: 1.9% (CO, $z = 0.71$), 6.9% (Hybrid, $z = 3.34$), 2.8% (OL+BDC, $z = 1.07$).

- Only the Hybrid loss is distinguishable from zero. CO and OL+BDC are within Monte Carlo error.
- At or below the 8-13% range earlier exploratory runs suggested.

Under covariance moderation the loss is much larger and strongly design-dependent, reaching 30.6% at OL+BDC but absent at CO.

- Relative loss: 3.0% (CO, not significant), 15.4% (Hybrid), 30.6% (OL+BDC), roughly 11 times the matched mean-moderation loss.
- The CO cell fluctuates within MC error, consistent with no architecture-by-carryover interaction there.
- Below the 40-60% range earlier runs suggested, but confirming substantial, design-dependent erosion.

3.2 3.2 Why the architectures diverge

Under mean moderation, carryover only compresses the treatment contrast, and the biomarker signal-to-noise ratio is nearly preserved.

- Carryover shrinks `Dbc` and inflates off-drug BR means (shared with covariance moderation).
- But β_{bm} is a fixed mean parameter with drug-state- independent noise.
- So the interaction variance falls but the signal-to-noise ratio degrades only modestly.

Covariance moderation's interaction signal is attacked from both sides, an attack that mean moderation escapes, which accounts for the divergence.

- The `bm:Dbc` coefficient depends on the covariance of B_i with `Dbc`-weighted outcomes.
- Under covariance moderation both the `Dbc` contrast and the BM-BR correlation decay together.
- Under mean moderation the second attack is absent, so power is largely preserved.

3.3 3.3 Robustness check at $N = 140$

An $N = 140$ confirmation run checks that the architecture ordering survives larger samples.

- Section 3.1 puts $N = 70$ on this design, where the loss is mechanically visible (no ceiling saturation).
- The check runs 12 cells at $N = 140$ (70 per path), 800 replicates each, on the two-path OL+BDC design.
- 100% convergence across 9,600 fits.

At $N = 140$ the ordering between the two architectures persists, though absolute magnitudes are compressed by ceiling saturation.

- Covariance moderation loses 9.3 points ($z = 7.4$), and mean moderation loses 1.7 points ($z = 2.0$).

- Both sit within 2-5 points of the ceiling at $t_{1/2} = 0$, truncating the absorbable loss.
- The Section 3.1 results remain the canonical reference for the narrative magnitude.

4 4. Which recipe matches which biology?

Use mean moderation when the biomarker mechanically sets the size of the drug effect.

- The biomarker fixes effective dose or receptor occupancy, so it scales the effect proportionally.
- Doubling the biomarker roughly doubles the effect at every timepoint.
- Carryover keeps the same proportion in the off-drug period, so power is barely touched.

Use covariance moderation when the biomarker is only a proxy for a hidden responder state.

- The biomarker imperfectly indexes the real cause, so identical biomarkers can respond differently.
- It predicts response only while the drug is acting.
- In the off-drug period the correlation fades and power drops sharply, as in Section 3.

The motivating prazosin/PTSD case is a covariance-moderation story, so its carryover power loss is a clinically relevant warning.

- Biomarker is resting blood pressure, a proxy for noradrenergic tone, not a driver of drug levels.
- Designs with long off-drug periods may lose more power than a mean-moderation simulation predicts.
- If blood pressure also acts through the mean channel, the dual-channel architecture of the companion paper applies.

5 5. Implications for Simulation Study Design

5.1 5.1 Choosing the appropriate architecture

When the mechanism is ambiguous, use the covariance moderation as the conservative default.

- Covariance moderation is the safer default for design decisions.
- Its larger carryover sensitivity keeps power estimates from being optimistic.
- A dual-channel sweep across mixing weights is available in the companion paper.

5.2 5.2 Reporting requirements

These requirements specialize the ADEMP template, and the architecture choice is its data-generating (D) element.

- ADEMP fixes Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.
- The DGP-architecture choice is the D element.
- Our reporting rules are ADEMP specialized to architecture-sensitive biomarker simulations.

6 6. Can a smarter analysis recover the lost power?

Covariance moderation's loss has two sources, and only the variance-compression part is recoverable, by down-weighting low-signal off-drug points.

- Source 1: the drug exposure variable's variance shrinks (recoverable).
- Source 2: the biomarker-response correlation genuinely fades off-drug (lost information).
- Methods that down-weight or drop off-drug points can help with the first.

7 7. How this fits the wider literature

Mean moderation is near-universal in the literature, so published power estimates may be optimistic for covariance moderation biomarkers.

- Simon, Freidlin-Korn, Renfro-Sargent, and standard N-of-1 studies all put the effect in the mean.
- Even N-of-1 carryover work sharing Hendrickson's authors uses mean moderation, so Hendrickson's choice looks like a one-off.
- Most published power figures may be optimistic for genuinely covariance moderation biomarkers.

The distinction matters most in treatment-switching designs. When the mechanism is unknown, compute power both ways and use the lower number.

- Treatment-switching designs (crossover, N-of-1, basket) put patients through off-drug phases where the covariance-moderation correlation decays.
- Parallel-group designs have no off-drug phase, so the architecture choice does not matter.
- This is a DGP question, separate from the analysis-model guidance of the PATH statement.

8 References